Diabetes Prediction Literature Review Draft

1. Literature Review

1.1 Classical Machine Learning and Population-Scale Risk Modeling

Machine learning for diabetes prediction has evolved from conventional classifiers on survey and clinical datasets toward more specialized models that explicitly address imbalance, missing values, and interpretability. A foundational large-scale study by Xie et al. used 138,146 participants from the 2014 Behavioral Risk Factor Surveillance System (BRFSS), including 20,467 individuals with type 2 diabetes, to compare support vector machines (SVM), decision trees, logistic regression, random forest, neural networks, and Gaussian Naive Bayes. Their results showed that all models achieved reasonable discrimination, with AUC values ranging from 0.7182 to 0.7949. The neural network delivered the highest accuracy (82.4%) and AUC (0.7949), while the decision tree yielded the highest sensitivity (51.6%), which is valuable for screening-oriented settings. This study remains important because it demonstrated that large, population-level survey data can support diabetes risk prediction even before more recent boosting and deep tabular models became common.

More recent work has expanded this line of research by integrating explainability into classical prediction pipelines. Ahmed et al. analyzed 253,680 BRFSS responses using logistic regression and random forest models with LIME and SHAP explanations. Their study reported approximately 86% test accuracy and emphasized that explainability methods can reveal how demographic, behavioral, and health-history variables drive model predictions. This is significant because diabetes prediction systems are increasingly expected not only to classify risk accurately, but also to provide clinicians with understandable reasons for the prediction. Together, Xie et al. and Ahmed et al. show that classical machine learning remains relevant when supported by large datasets and interpretable post-hoc analysis.

Another important methodological contribution comes from Roy et al., who focused on missing values and class imbalance in type 2 diabetes classification. Their framework compared multiple imputation strategies, including median, KNN, and iterative imputation, and then balanced the best-performing dataset with SMOTETomek before training an artificial neural network. The final model achieved 98% accuracy on the test set. Although this paper is older than several of the other studies reviewed here, it directly addresses a recurring problem in diabetes datasets: incomplete records combined with skewed class distributions. This makes it highly relevant for projects that rely on practical clinical or survey data rather than carefully curated benchmark datasets.

1.2 Gradient Boosting and Tree-Based Ensemble Models

Recent diabetes prediction research strongly favors gradient boosting and tree-based ensemble methods because they handle nonlinear feature interactions well and often require less feature engineering than traditional approaches. In a treatment recommendation context, Alqrinawi et al. developed a CatBoost-based model using 51,000 anonymized patient records from the Palestinian Ministry of Health. Their CatBoost model reached 97% accuracy and outperformed baseline approaches such as random forest and SVM. A major advantage of CatBoost in this setting is its native handling of categorical variables, which is particularly useful in diabetes records containing demographic, medication, and clinical history fields.

A second CatBoost-oriented study by Sumathi et al. compared decision trees, random forest, KNN, Naive Bayes, AdaBoost, XGBoost, multilayer perceptron, and ensemble combinations for diabetes prediction. Their results highlighted CatBoost's computational efficiency: the proposed CatBoost model completed execution in 4.27 seconds, compared with 314.96 seconds for an ensemble alternative, making it approximately 98.64% faster. This finding is important because model selection in healthcare is not only about predictive performance, but also about deployment feasibility, especially when systems must run quickly on large or continuously updated datasets.

Other boosting-based approaches reinforce the same pattern. Khurshid et al. optimized XGBoost with Bayesian hyperparameter tuning and reported 97.26% accuracy, 95.72% F1-score, and an MCC of 81.18, slightly improving on grid-search-based XGBoost. Similarly, Shao et al. combined class balancing with Optuna-tuned LightGBM on a 100,000-record diabetes dataset and improved accuracy from 97.07% to 97.11% while raising precision from

97.17% to 98.99%; they also reported a single search time of only 2.5 seconds. These results collectively suggest that modern boosting methods are particularly strong candidates for tabular diabetes prediction because they balance predictive strength, scalability, and training efficiency.

Abousaber et al. broadened this evidence by testing multiple machine learning models under several imbalance-handling methods across the PIMA, Diabetes Dataset 2019, and BIT_2019 datasets. On the Diabetes Dataset 2019 benchmark, KNN with ADASYN achieved 97.25% accuracy, 95.04% F1-score, 98.53% specificity, and 0.9952 ROC-AUC. On the PIMA dataset, random forest, XGBoost, and LightGBM paired with oversampling methods achieved near-perfect or perfect reported performance. Their study is valuable because it shows that strong results often come not from a single model alone, but from the interaction between model family and class-balancing strategy.

1.3 Imbalance Handling and Synthetic Data Augmentation

Class imbalance remains one of the central technical issues in diabetes prediction. In many datasets, non-diabetic cases outnumber diabetic cases, causing standard classifiers to favor majority-class accuracy at the expense of minority-class detection. Netayawijit et al. addressed this directly by integrating SMOTE with SHAP in a seven-stage pipeline and evaluating Random Forest, Gradient Boosting, SVM, Logistic Regression, and XGBoost on a stratified sample of 1,500 records from a 100,000-record diabetes dataset. Their Random Forest-SMOTE model achieved 96.91% accuracy, AUC of 0.998, sensitivity of 99.5%, and specificity of 97.3%. The study is especially useful because it combines strong predictive performance with rigorous validation and interpretable outputs.

Synthetic data generation has also emerged as an alternative to interpolation-based oversampling. Nematollahi et al. used a Conditional Tabular GAN (CTGAN) to augment body-composition data before classification with a multilayer perceptron. Their CTGAN-MLP approach achieved 93.91% accuracy, 93.87% AUC, 94.48% precision, and 93.89% F1-score under stratified 5-fold cross-validation. Unlike standard oversampling approaches, CTGAN attempts to learn the underlying data distribution, which may better preserve nonlinear relationships among variables. This is particularly relevant in diabetes prediction because features such as fat percentage, basal metabolic rate, and fat-free mass often interact in complex ways.

Shao et al. and Roy et al. also fit within this theme, since both demonstrate that balancing and preprocessing can materially change downstream performance. The literature therefore suggests that imbalance handling should not be treated as a minor preprocessing step; rather, it is a core design decision that shapes whether diabetes prediction models are clinically useful, especially when the goal is to identify high-risk patients rather than maximize overall accuracy alone.

1.4 Deep Learning for Structured Diabetes Prediction

Although deep learning is traditionally associated with images and text, recent research has adapted it for structured diabetes datasets. Rahardi et al. compared two advanced deep tabular architectures, TabNet and Neural Oblivious Decision Ensembles (NODE), and then combined them through a soft-voting ensemble. Their final ensemble achieved 93.55% validation accuracy, 92.60% precision, 94.58% recall, and 93.58% F1-score. The value of this study lies in showing that attention-based and neural-oblivious architectures can compete effectively on tabular diabetes data when proper preprocessing and oversampling are applied.

Farnoosh et al. proposed DiabetesXpertNet, an attention-based CNN specifically designed for tabular diabetes prediction rather than image inputs. Evaluated on the PIMA Indian Diabetes dataset and the Frankfurt Hospital dataset, the model achieved 89.98% accuracy, 91.95% AUC, 89.08% precision, 88.11% recall, and 88.01% F1-score. The architecture uses dynamic channel attention and context-aware feature enhancement to prioritize clinically meaningful variables such as glucose and insulin. This makes the study notable because it shows how deep models can be adapted to preserve the structure of tabular medical data without converting them into artificial image-like representations.

Nematollahi et al.'s CTGAN-MLP framework also belongs partly in this category because it couples generative modeling with a neural predictor. Taken together, these studies indicate that deep learning can outperform or complement classical models in diabetes prediction, but its success depends heavily on architecture design, preprocessing quality, and the availability of sufficiently informative structured data.

1.5 Explainability and Clinical Trust

Explainability is now a major requirement in diabetes-related machine learning because clinicians need transparent justification before using model outputs in screening or decision support. Ahmed et al. explicitly compared LIME and SHAP for diabetes prediction explanations and showed how both local and global explanations can clarify the behavior of logistic regression and random forest models on large BRFSS data. Their work is useful not merely because it adds interpretability, but because it frames explanation quality as part of model evaluation rather than as an afterthought.

Netayawijit et al. extended this direction by integrating SHAP into a class-balanced prediction framework and showing that glucose and BMI were the most influential variables. This is consistent with known clinical risk factors and demonstrates how explainability can improve trust in model outputs. Likewise, Nematollahi et al. used SHAP to identify important body-composition variables, while Farnoosh et al. emphasized interpretability in the design of DiabetesXpertNet by focusing attention on clinically relevant features.

Despite these advances, many high-performing diabetes studies still prioritize accuracy and AUC without providing clinically meaningful explanations, calibration analysis, or external validation. This creates a persistent gap between algorithmic performance in research settings and practical adoption in clinical workflows.

2. Research Gap

The reviewed literature shows substantial progress in diabetes prediction, but several major gaps remain. First, many studies report very high performance on public or derived benchmark datasets such as PIMA or Kaggle-hosted diabetes datasets, yet comparatively fewer studies validate their methods on real, heterogeneous clinical populations. This raises concerns about generalizability across hospitals, demographics, and healthcare systems.

Second, class imbalance is widely recognized, but the handling strategies vary considerably across studies. Some papers use SMOTE, ADASYN, or hybrid resampling, while others rely on generative models such as CTGAN. Because these methods are often evaluated on different datasets under different validation protocols, it is still unclear which balancing strategy is most reliable for real-world diabetes screening tasks.

Third, there is no consistent agreement on the best model family. Classical methods such as logistic regression and decision trees remain valuable for interpretability and sensitivity, whereas CatBoost, XGBoost, and LightGBM often offer stronger overall predictive performance, and deep tabular architectures such as TabNet, NODE, and attention-based CNNs promise further gains. However, few studies compare these families under a unified preprocessing, balancing, and explainability pipeline.

Fourth, explainability is still unevenly integrated. Some papers treat SHAP or LIME as a central design component, while others focus almost entirely on accuracy. For clinical deployment, strong discrimination alone is insufficient; models should also provide stable explanations, reveal clinically plausible feature importance, and support patient-level interpretation.

Based on these gaps, a strong project contribution would be to build and compare several diabetes prediction models under one consistent workflow that includes: rigorous preprocessing, explicit imbalance handling, evaluation with multiple metrics beyond accuracy, and explainability analysis using methods such as SHAP or LIME. Such a study would contribute more than another isolated accuracy report because it would identify which models remain strong when fairness, interpretability, and practical usability are considered together.

3. Comparative Analysis of Reviewed Methods

Reference	Algorithm / Focus	Dataset	Best Reported Result	XAI	Main Limitation
Xie et al. (2019)	SVM, DT, LR, RF, NN, GNB	BRFSS 2014 (138,146 participants)	NN: 82.4% accuracy, AUC 0.7949; DT highest sensitivity 51.6%	No explicit XAI	Moderate predictive power; survey data only
Ahmed et al. (2025)	LR and RF with LIME/SHAP	BRFSS 2015 (253,680 responses)	About 86% test accuracy	LIME, SHAP	Explainability strong, but model family remains relatively simple
Roy et al. (2021)	Imputation + SMOTETomek + ANN	Type 2 diabetes tabular dataset	98% test accuracy	No	Single-dataset framing; limited discussion of external validation
Alqrinawi et al. (2025)	CatBoost for treatment prediction	Palestinian Ministry of Health records (51,000 patients)	97% accuracy	No	Focuses on treatment recommendation rather than general diabetes onset prediction
Sumathi et al. (2025)	CatBoost comparative study	Diabetes prediction benchmark data	CatBoost run time 4.27 s vs. 314.96 s ensemble baseline	No	Emphasis is stronger on computational efficiency than interpretability
Shao et al. (2024)	LightGBM + Optuna + SMOTE/RUS	Kaggle diabetes dataset (100,000 records)	97.11% accuracy, 98.99% precision	No	Needs external validation beyond benchmark data
Khurshid et al. (2025)	XGBoost + Bayesian optimization	Public Kaggle diabetes dataset	97.26% accuracy, 95.72% F1-score, MCC 81.18	No	Gains over grid search are modest
Netayawijit et al. (2025)	RF, GB, SVM, LR, XGB with SMOTE	1,500-sample subset from 100,000-record dataset	RF-SMOTE: 96.91% accuracy, AUC 0.998	SHAP	Proof-of-concept scale; subset-based evaluation
Rahardi et al. (2025)	TabNet + NODE ensemble	Diabetes tabular dataset with oversampling	93.55% accuracy, 93.58% F1-score	No	Limited clinical interpretability analysis
Nematollahi et al. (2025)	CTGAN + MLP	Body composition dataset	93.91% accuracy, AUC 93.87%	SHAP	Synthetic-data dependence may affect external realism
Farnoosh et al. (2025)	DiabetesXpertNet attention-based CNN	PIMA + Frankfurt Hospital datasets	89.98% accuracy, AUC 91.95%	Attention-based interpretability	Deep model complexity may limit straightforward deployment
Abousaber et al. (2025)	Multi-model imbalanced-learning framework	PIMA, Diabetes 2019, BIT_2019	Diabetes 2019: KNN + ADASYN 97.25% accuracy, ROC-AUC 0.9952	No	Results vary strongly by dataset and resampling method

References

1. Z. Xie, O. Nikolayeva, J. Luo, and D. Li, “Building Risk Prediction Models for Type 2 Diabetes Using Machine Learning Techniques,” *Preventing Chronic Disease*, vol. 16, 2019, Art. 190109, doi: 10.5888/pcd16.190109.

2. S. Ahmed, M. S. Kaiser, M. S. Hossain, and K. Andersson, “A Comparative Analysis of LIME and SHAP Interpreters With Explainable ML-Based Diabetes Predictions,” *IEEE Access*, vol. 13, 2025, doi: 10.1109/ACCESS.2024.3422319.

3. K. Roy, M. Ahmad, K. Waqar, K. Priyaah, J. Nebhen, S. S. Alshamrani, M. A. Raza, I. Ali, and M. I. Uddin, “An Enhanced Machine Learning Framework for Type 2 Diabetes Classification Using Imbalanced Data with Missing Values,” *Complexity*, vol. 2021, 2021, Art. 9953314, doi: 10.1155/2021/9953314.

4. S. S. M. Alqrinawi, M. A. Burhanuddin, and L. Salahuddin, “CatBoost Model for Enhanced Treatment Prediction in Type 2 Diabetes Patients,” *Journal of Information Systems Engineering & Management*, vol. 10, no. 52s, 2025, doi: 10.52783/jisem.v10i52s.10680.

5. M. Sumathi, S. Sahana, S. Sri Raja Rajeswari, V. Kruthi, and S. P. Raja, “Enhancing Diabetes Prediction With Minimal Processing Time Using CatBoost: A Comparative Study,” *Journal of Evaluation in Clinical Practice*, vol. 31, no. 7, 2025, doi: 10.1111/jep.70284.

6. H. Shao, X. Liu, D. Zong, et al., “Optimization of diabetes prediction methods based on combinatorial balancing algorithm,” *Nutrition & Diabetes*, vol. 14, 2024, Art. 63, doi: 10.1038/s41387-024-00324-z.

7. M. R. Khurshid, S. Manzoor, T. Sadiq, L. Hussain, M. S. Khan, and A. K. Dutta, “Unveiling diabetes onset: Optimized XGBoost with Bayesian optimization for enhanced prediction,” *PLOS ONE*, vol. 20, no. 1, 2025, e0310218, doi: 10.1371/journal.pone.0310218.

8. P. Netayawijit, W. Chansanam, and K. Sorn-In, “Interpretable Machine Learning Framework for Diabetes Prediction: Integrating SMOTE Balancing with SHAP Explainability for Clinical Decision Support,” *Healthcare*, vol. 13, no. 20, 2025, Art. 2588, doi: 10.3390/healthcare13202588.

9. M. Rahardi, F. F. Abdulloh, A. Masruro, B. P. Asaddulloh, A. Aminuddin, and N. Sholihah, “Ensemble Learning for Diabetes Prediction: An Integration of TabNet and Neural Oblivious Decision Ensembles (NODE),” *Engineering, Technology & Applied Science Research*, vol. 15, no. 6, pp. 30426-30431, 2025, doi: 10.48084/etasr.15192.

10. M. A. Nematollahi et al., “Enhanced diabetes prediction using CTGAN-MLP approach on body composition data,” *Scientific Reports*, vol. 16, Art. 2134, 2026, doi: 10.1038/s41598-025-31928-9. Published online Dec. 10, 2025.

11. R. Farnoosh, K. Abnoosian, R. A. Isewid, and D. Javaheri, “DiabetesXpertNet: An innovative attention-based CNN for accurate type 2 diabetes prediction,” *PLOS ONE*, vol. 20, no. 9, 2025, e0330454, doi: 10.1371/journal.pone.0330454.

12. I. Abousaber, H. F. Abdallah, and H. El-Ghaish, “Robust predictive framework for diabetes classification using optimized machine learning on imbalanced datasets,” *Frontiers in Artificial Intelligence*, vol. 7, 2025, Art. 1499530, doi: 10.3389/frai.2024.1499530.